

Labeling Guideline

Introduction

To verify the quality of the dataset, we conduct a manual analysis in a subset of the **Kernel Fixing Commit (KFC)** dataset. This analysis aims to confirm that the "Fixes: " field is indeed used in practice to point to the cause of the bug (as opposed to pointing to the origins of the functionalities involved in a future bug). Since in some cases, the developer may point to the commit that added the functionality that was later involved in a bug-fix, with the actual bug being manifested or introduced in the meantime.

In this study, four steps outline the process to validate the quality of a subset of the **KFC** dataset:

1. **Sampling:** The **KFC** dataset comprises 76,046 bug-fixing commits. A random sample of 383 cases is selected to ensure a statistically representative analysis. This sample size is determined using a widely accepted sample size calculator [1], targeting a 95% confidence level and a 5% margin of error.
2. **Preparation:** Before labeling, three annotators engage in discussions to refine their approach. This collaborative effort, based on an initial examination of a small batch of cases, leads to the development of a set of labeling guidelines.
3. **Labeling:** Adhering to these guidelines, the annotators proceed to label the dataset. One annotator is responsible for all 383 cases, while the others split the remaining cases, labeling 191 and 192 cases, respectively.
4. **Discussion:** After labeling, the annotators convene to address any discrepancies. During these discussions, cases of disagreement are thoroughly examined. For instances where consensus is elusive, insights from Linux experts are sought to inform their final decisions.

[1] <https://www.surveymonkey.com/mp/sample-size-calculator>

Guideline Details

During the preparation phase for labeling the **Kernel Fixing Commit (KFC)** dataset, discussions revealed that not all cases directly point to the cause of the bug. Specifically, the annotators identified the following scenarios:

1. **Affected by Other Changes:** For certain bug-fixing commits, such as commit f7fc391a, the initially labeled bug-inducing commit was correct at the time of its creation. However, subsequent changes by other commits led to the manifestation of the bug. These cases do not directly represent the bug's cause and **should be labeled as "Affected by other changes."**
2. **Partial Fix:** Illustrated in Figure 1, this scenario involves a commit that introduces buggy code and a subsequent commit that attempts but fails to fully resolve the bug, followed by another commit that completely fixes the bug. The labeling criteria for this scenario are as follows:

- a. If the final bug-fixing commit (i.e., one more fix in Fig 1) correctly identifies the original buggy code commit, it points directly to the bug's cause (i.e., introduces buggy code in Fig 1). **This should be labeled as "yes" and marked as "partial fix - a."**
- b. If the final bug-fixing commit labels the intermediate, unsuccessful fix attempt as the bug-inducing commit (i.e., partial fix in Fig 1), which did not introduce new bugs but failed to fully correct the original issue, **it should be labeled as "Partial fix."**
- c. If the final bug-fixing commit identifies an intermediate fix attempt that unintentionally introduced another bug, leading to the need for further fixes, **it should be labeled as "yes" and marked as "partial fix - c."**

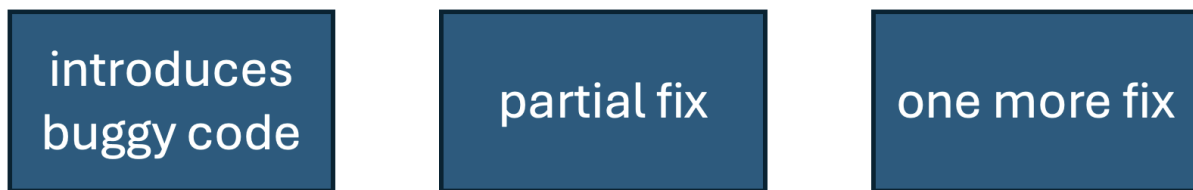


Fig 1. Partial Fixing Example

3. **Introducing a New Function:** In cases where a commit introduces a new function and the function undergoes multiple changes before the bug-fixing commit, **without clear evidence** that the initial commit directly caused the bug, yet the developer labeled it as the bug-inducing commit, **these should be labeled as "Introducing a new function."**

Summary

Therefore, annotators are guided to use the following categories for labeling:

1. "yes": The bug-inducing commit clearly points to the cause of the bug.
2. "Affected by other changes": External modifications led to the bug's emergence.
3. "Partial fix": An intermediate commit attempted to fix the bug but did not completely succeed.
4. "Introducing a new function": The commit introduced a function that was later involved in a bug, without direct causation.
5. "no": The case does not point to the cause of the bug and does not fit into categories 2-4. **The annotator should mark the reason for this situation.**
6. "not sure": The annotator is uncertain whether the case points to the cause of the bug. **The annotator should mark the reason for this situation.**

When dealing with partial fix situations, even if the labeled bug-inducing commit represents the cause of the bug, annotators should also mark "partial fix -a" or "partial fix -c" after labeling it as "yes."